



MID YEAR PROJECT

2083

TRAILBLAZERS - BOOK 1



ABOUT MID-YEAR PROJECT

The Mid-Year Project is a cornerstone of the Digi Curriculum, mandatory for all enrolled students. It serves as a primary measure of development during Phase I, designed to move students beyond theoretical understanding into practical application. With a weight of 50 Marks, it commands a significant 35% share in the Annual Digi Evaluation, underscoring its importance in the Digi educational journey. Beyond individual grading, the Mid-Year Project functions as a nationwide competition among all Digischool-enrolled institutions.

MARKING SCHEME

Full Marks: **50 Marks**

Threshold Marks: **20 Marks**

Annual Evaluation Weightage: **35%**

ASSIGNMENT DESCRIPTION

TASK

Write pseudocode for an **algorithm** and draw a **flowchart** for **ONE** of the following real-life problems. Your solution must include a **sequence**, at least one **selection (IF/ELSE)**, and at least one **loop (REPEAT/WHILE)**:

- Smart Traffic Light System
- Library Book Borrowing System
- Canteen Ordering System
- Automatic Attendance Checker
- Online Quiz Grading System

Use a tool like Draw.io or *MS Word* to ease your flowchart creation process.

GUIDELINES

- Write a complete **pseudocode algorithm** for your chosen problem that clearly shows sequence, selection (IF/ELSE), and at least one loop (REPEAT/WHILE/FOR). Use correct indentation to highlight decisions and loops. **(15 marks)**
- Draw a **flowchart** that represents the same algorithm using correct flowchart symbols (oval for Start/End, parallelogram for Input/Output, rectangle for Process, diamond for Decision) and directional arrows showing the flow. **(15 marks)**
- Write a brief **description (50–100 words)** explaining how your algorithm works, what inputs it takes, what decisions it makes, and what output it produces. **(10 marks)**
- Identify and name **at least two variables** used in your algorithm, state their **data types** (e.g., Integer, String, Boolean), and explain what each stores. **(5 marks)**
- Demonstrate **creativity and originality** in the way you have approached the problem, including any unique logic, meaningful variable names, or creative extensions beyond the basic requirement. **(5 marks)**

FILES TO BE SUBMITTED

- **.PDF file** containing the pseudocode, flowchart, and written description

WHAT NOT TO DO

- Strict prohibition against plagiarism is in effect, and any project identified as a direct replication of another's work will be subject to rejection.
- Inclusion of images, language, or themes that could be deemed hurtful, disrespectful, or offensive to others is strongly discouraged in your project.
- Emphasize reliance on personal understanding and creativity, utilizing AI as a supplementary resource. Avoiding excessive dependence on AI is encouraged to preserve the uniqueness of your input.
- Adherence to project guidelines is strictly mandated. Deviation from these guidelines may have an impact on your assessment. If in doubt, seek clarification to ensure alignment of your work with project requirements.

- Do not submit a flowchart without its matching pseudocode, or pseudocode without its matching flowchart. Both parts are required and must represent the same algorithm.
- Accurate citation and referencing of all sources used in your project are expected to acknowledge the intellectual property of others. Failure to do so may have a negative impact on your assessment.
- Adherence to project deadlines and submission timelines is required. Late submissions may be subject to penalties or a reduction in the evaluation score.

MARKING RUBRICS

PSEUDOCODE QUALITY (15 MARKS)

0 – 5	6 – 10	11 – 15
Pseudocode is incomplete or missing. Sequence, selection, and loop are absent or incorrectly represented. No use of START/END or proper keywords.	Pseudocode is mostly complete. At least two of the three constructs (sequence, selection, loop) are present but may contain minor errors or missing indentation in some places.	Pseudocode is complete and correct. All three constructs (sequence, selection, loop) are present and clearly written with proper keywords, indentation, and START/END markers.

FLOWCHART ACCURACY (15 MARKS)

0 – 5	6 – 10	11 – 15
The flowchart is missing, incomplete, or uses incorrect symbols throughout. Arrows are missing or point in the wrong direction. No clear Start or End.	Most flowchart symbols are correct. Start/End present. Some minor symbol errors or missing arrow labels on decision branches (Yes/No).	The flowchart is accurate and complete. All four symbol types are used correctly. Decision branches clearly labelled (Yes/No). Arrows show correct flow including any loops.

WRITTEN DESCRIPTION (10 MARKS)

0 – 3	4 – 6	7 – 10
Description is missing or does not meaningfully explain the algorithm. Key computing terms are absent or used incorrectly.	Description partially explains the algorithm. Inputs, outputs, or decisions are mentioned, but not all are clearly described. Some computing terms are used correctly.	Description clearly explains how the algorithm works, identifies inputs and outputs, and uses correct computing terminology (e.g., sequence, selection, loop, variable).

VARIABLES AND DATA TYPES (5 MARKS)

0 – 1	2 – 3	4 – 5
Variables are missing or unnamed. No data types are identified. Variable names are unclear or inappropriate.	At least one variable is identified with a reasonable name and data type. Explanations may be partially complete or slightly inaccurate.	At least two variables are clearly named, correctly typed (e.g., Integer, String, Boolean), and well explained. Variable names are meaningful and appropriate.

CREATIVITY AND ORIGINALITY (5 MARKS)

0 – 1	2 – 3	4 – 5
The solution follows a basic, predictable approach with no individual thought. The algorithm solves the minimum of the problem with no additions.	Some creativity is shown, such as meaningful variable names, additional edge-case handling, or a thoughtfully structured algorithm that goes slightly beyond the minimum.	Exceptional creativity is demonstrated. The algorithm introduces additional features, clever logic, or real-world extensions that make the solution stand out and show deep understanding.